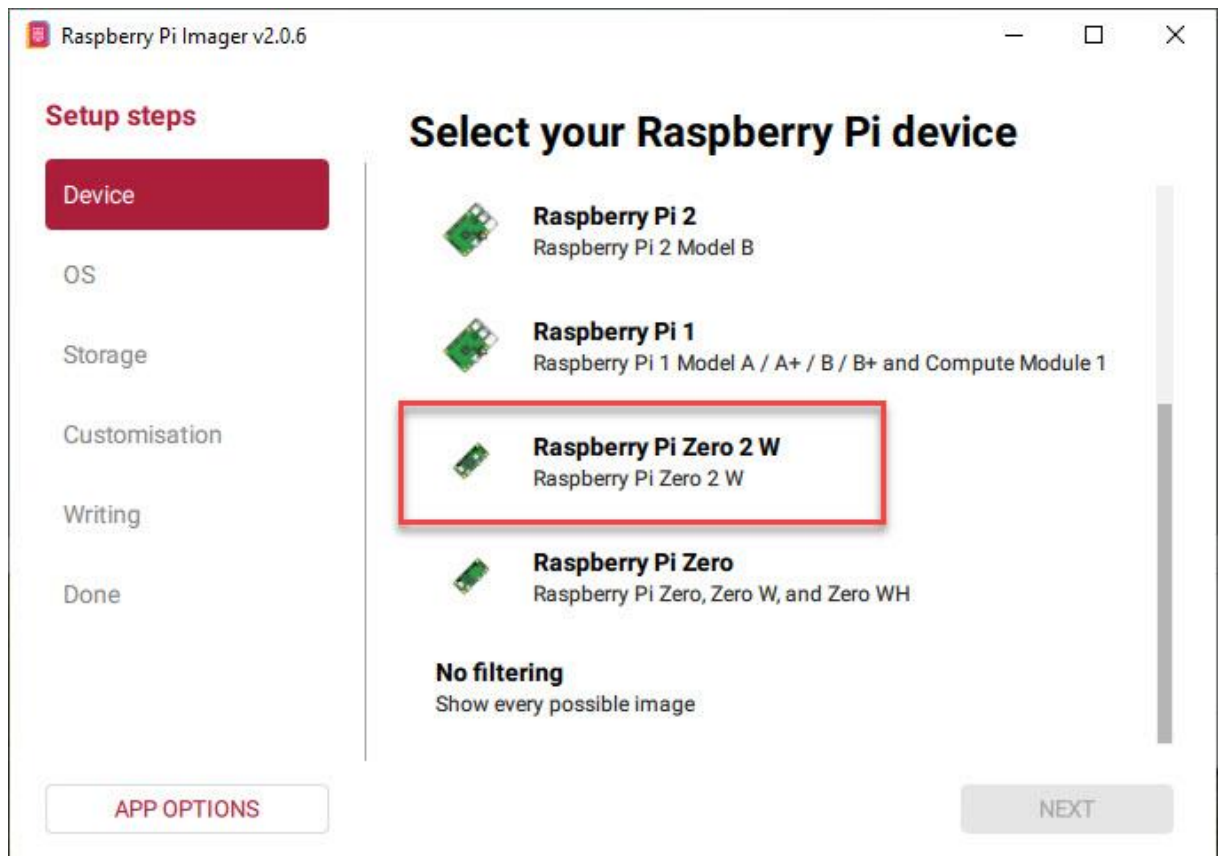


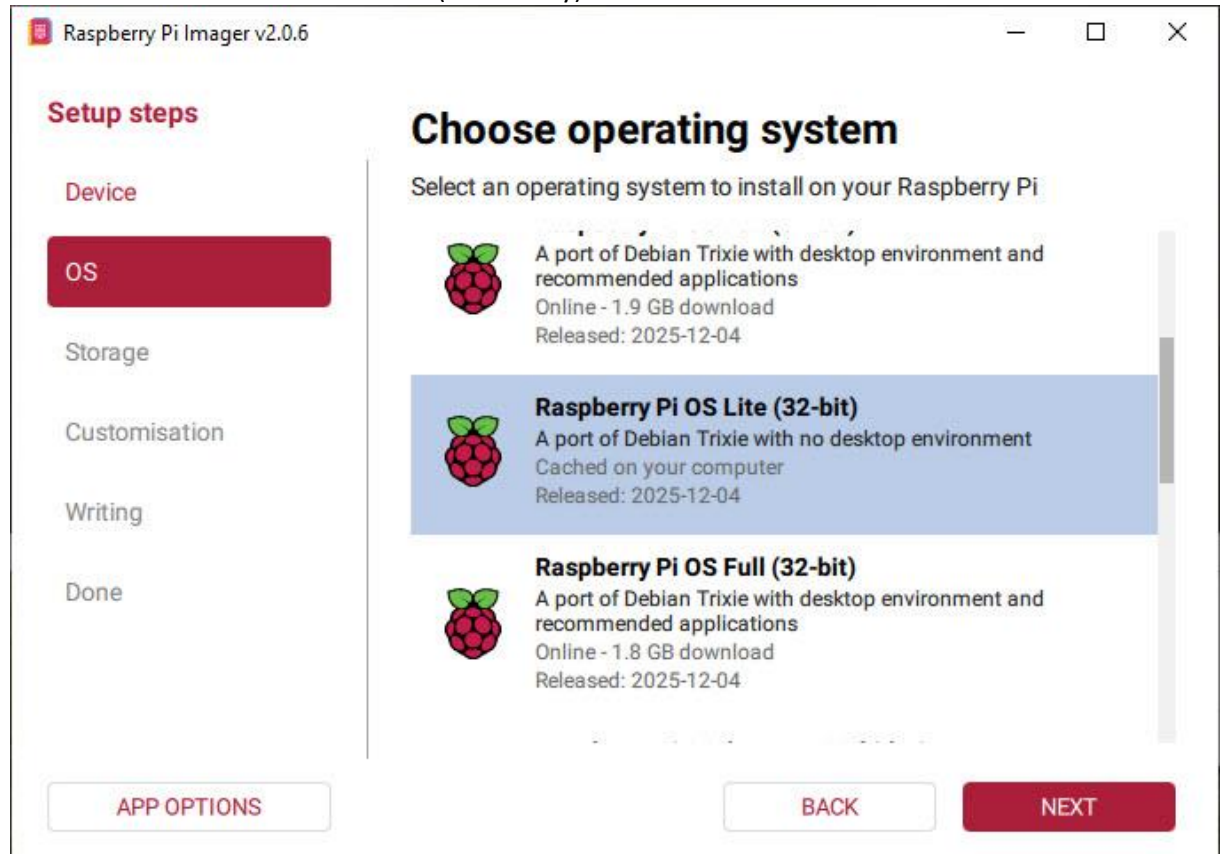
Pi-Ager install on Raspberry Pi 3/Pi 4/Pi 5/Pi zero (2)w under Pi OS 12 Lite Bookworm/Trixie

- **For all Pi:**
Download and install Raspberry Pi Imager v2.0.6 or later from <https://www.raspberrypi.com/software/>.

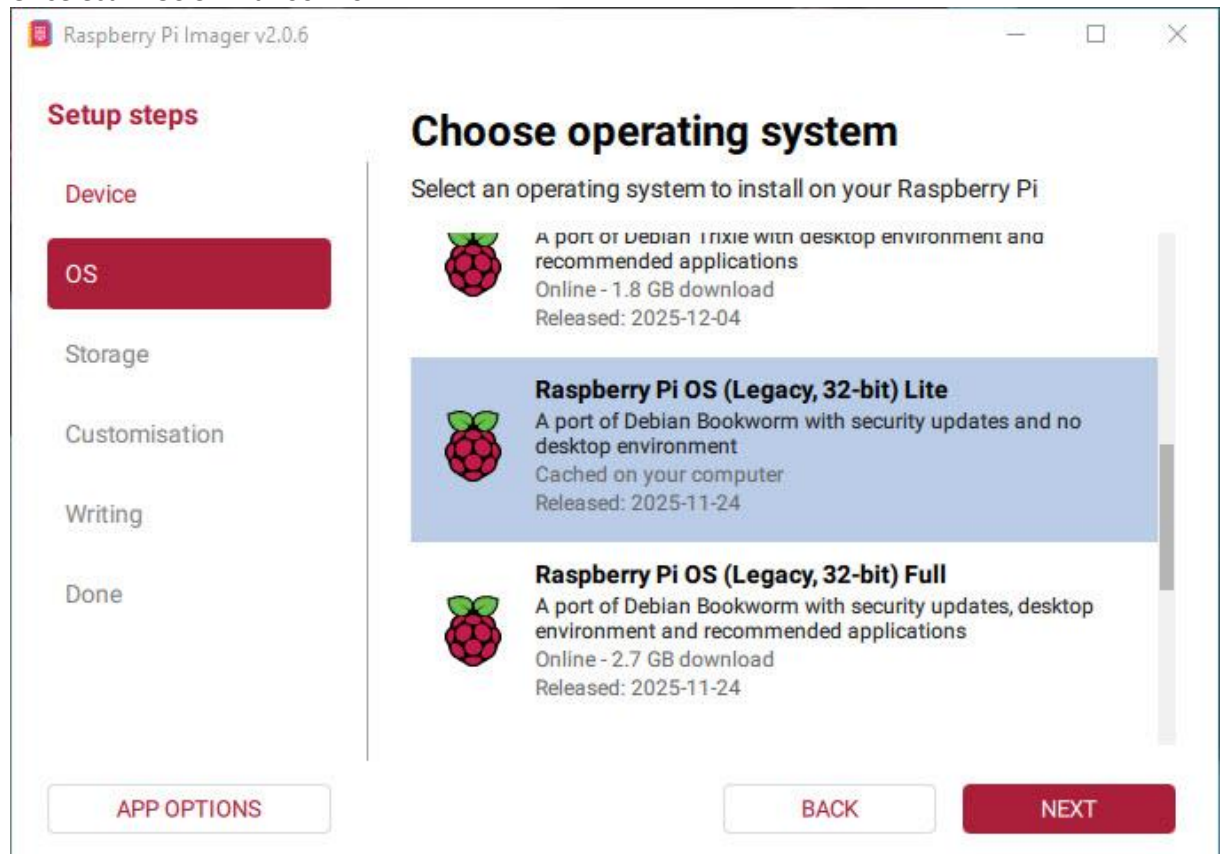
Start Raspberry Pi Imager, then continue with the follow steps:



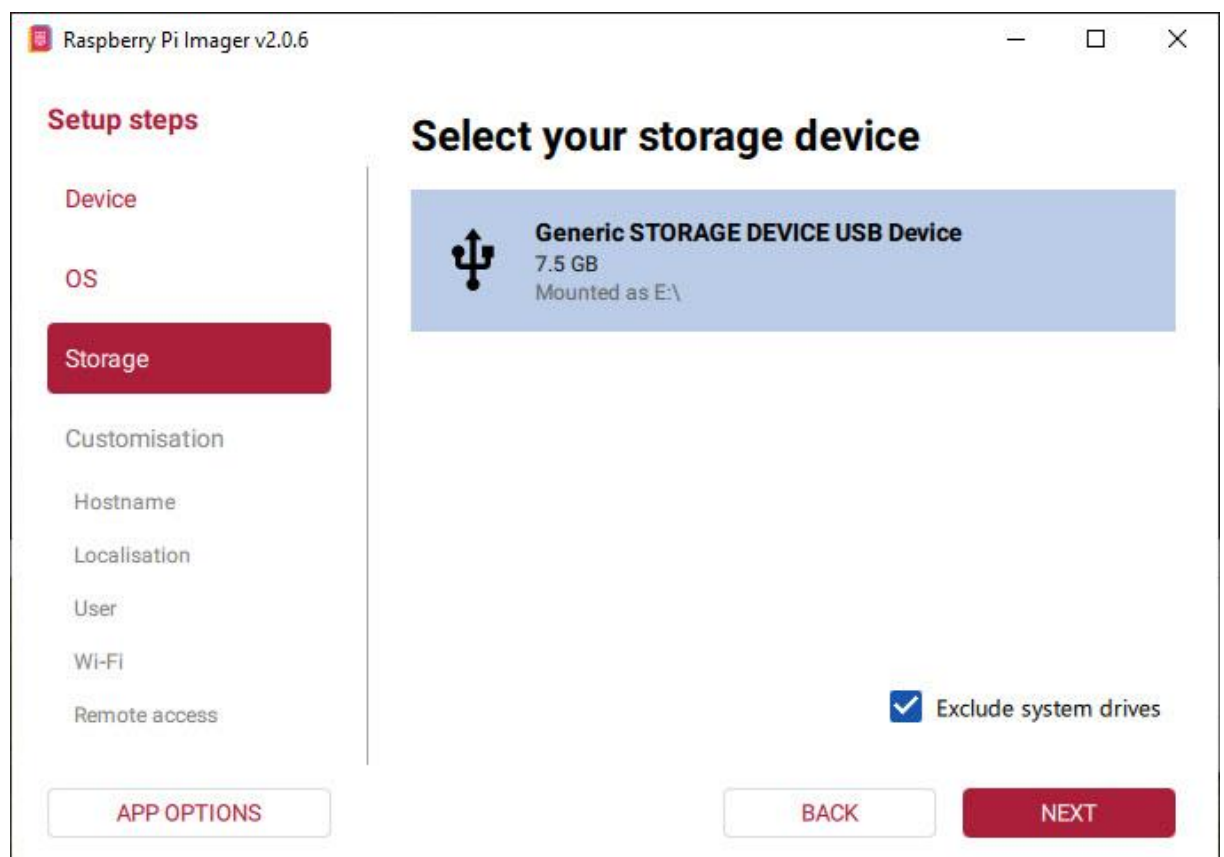
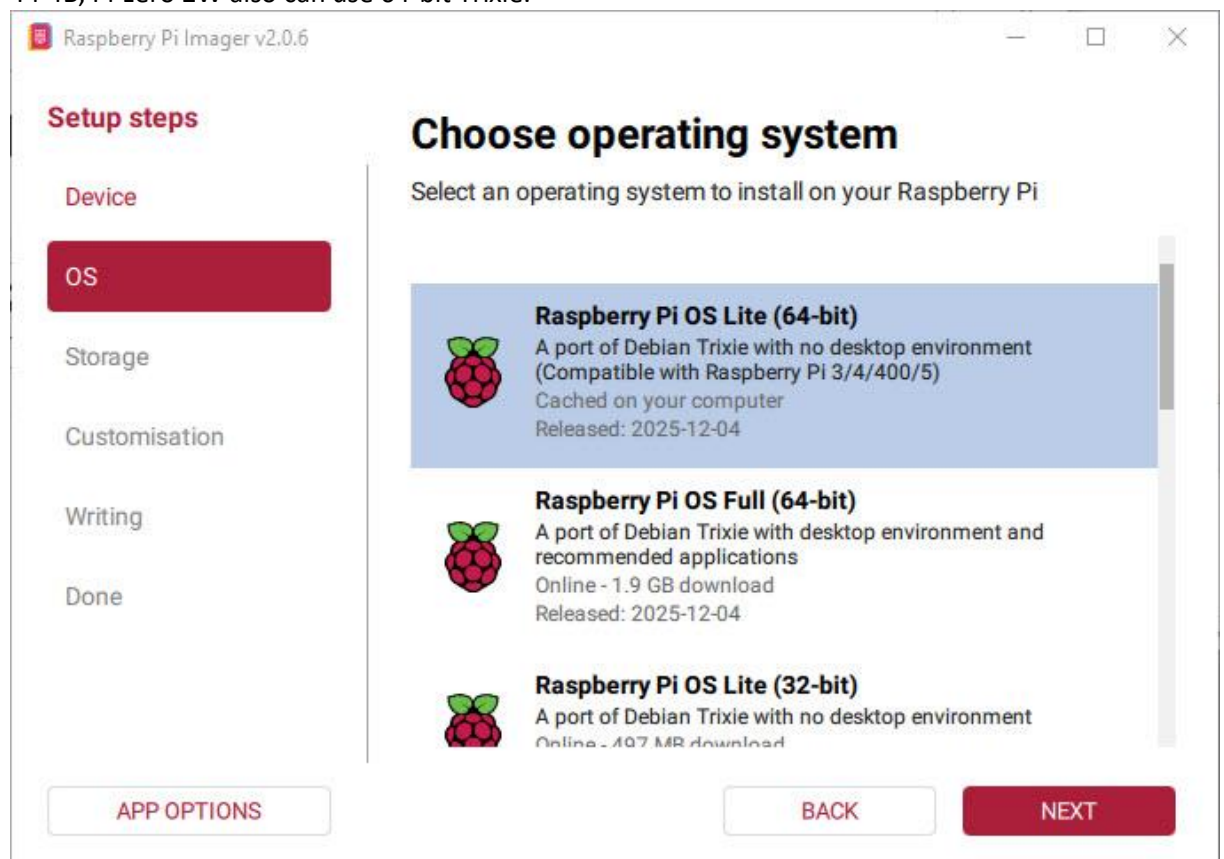
Select Pi-OS 32 Bit Trixie for Pi Zero (32-bit only)



Or select Pi-OS 32 Bit Bookworm:



Or if you want to install pi-ager on Raspberry Pi-5 that needs a 64 Bit PI-OS, select Trixie.
Pi-4B, Pi-zero 2W also can use 64-bit Trixie.



Raspberry Pi Imager v2.0.6

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Customisation: Choose hostname

Enter your hostname
RaspberryPi

A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Imager v2.0.6

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Customisation: Localisation

Select your location for suggested time zone and keyboard layout

Capital city: Berlin (Germany)

Time zone: Europe/Berlin

Keyboard layout: de

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Imager v2.0.6

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose username

Create a user account for your Raspberry Pi

Username:

Enter your username

pi

Password:

Saved (hidden) — leave blank to keep

••••••••

👁

Confirm password:

Re-enter to change password

••••••••

👁

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Imager v2.0.6

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose Wi-Fi

SECURE NETWORK

OPEN NETWORK

SSID:

Network name

••••••••••

Password:

Network password

••••••••••••••

👁

Confirm password:

Re-enter password

••••••••••••••

👁

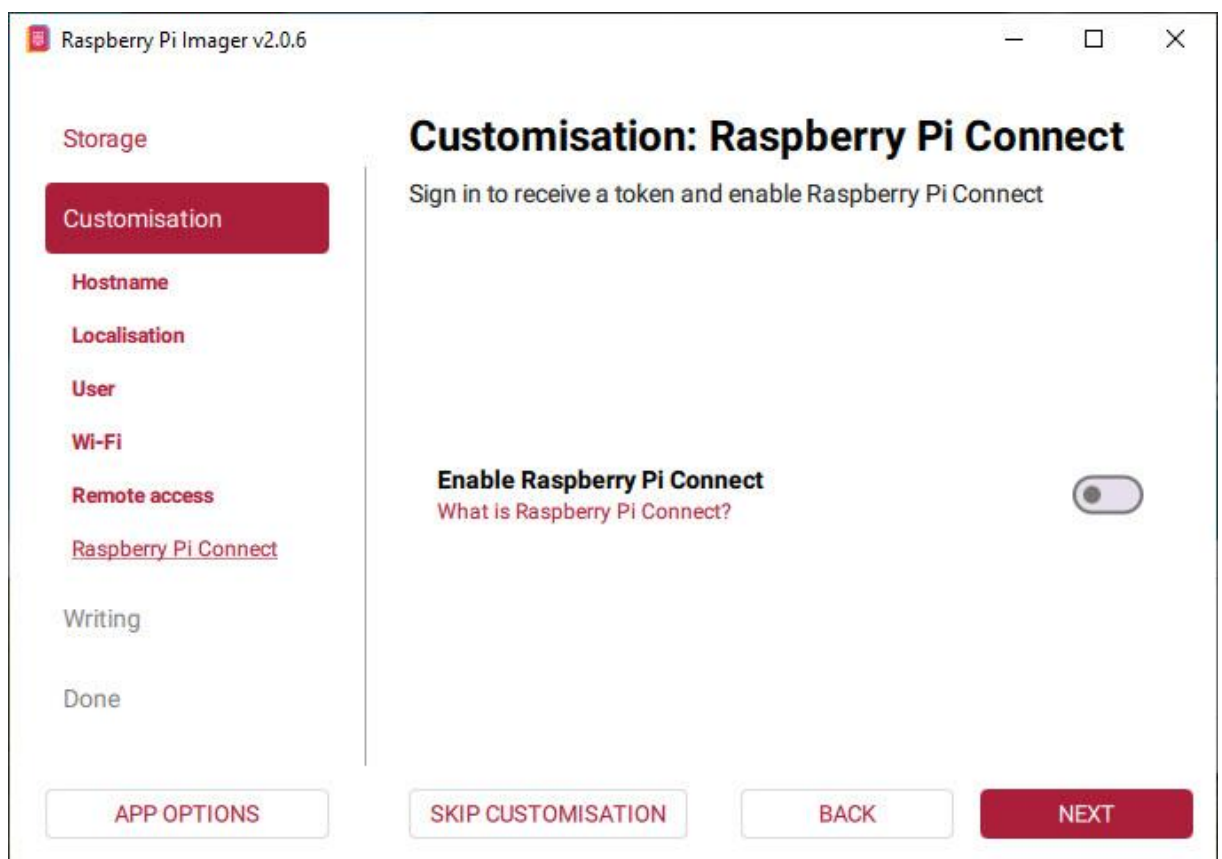
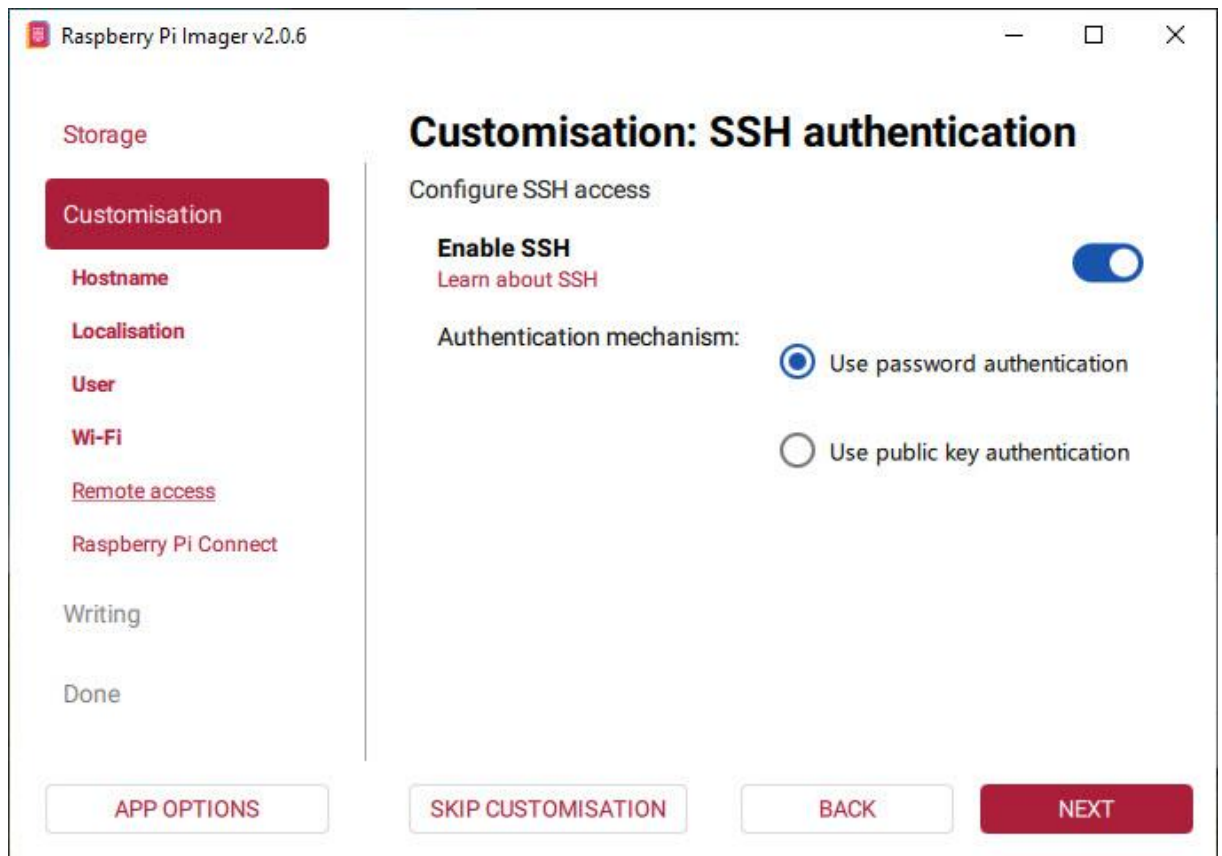
☐ Hidden SSID

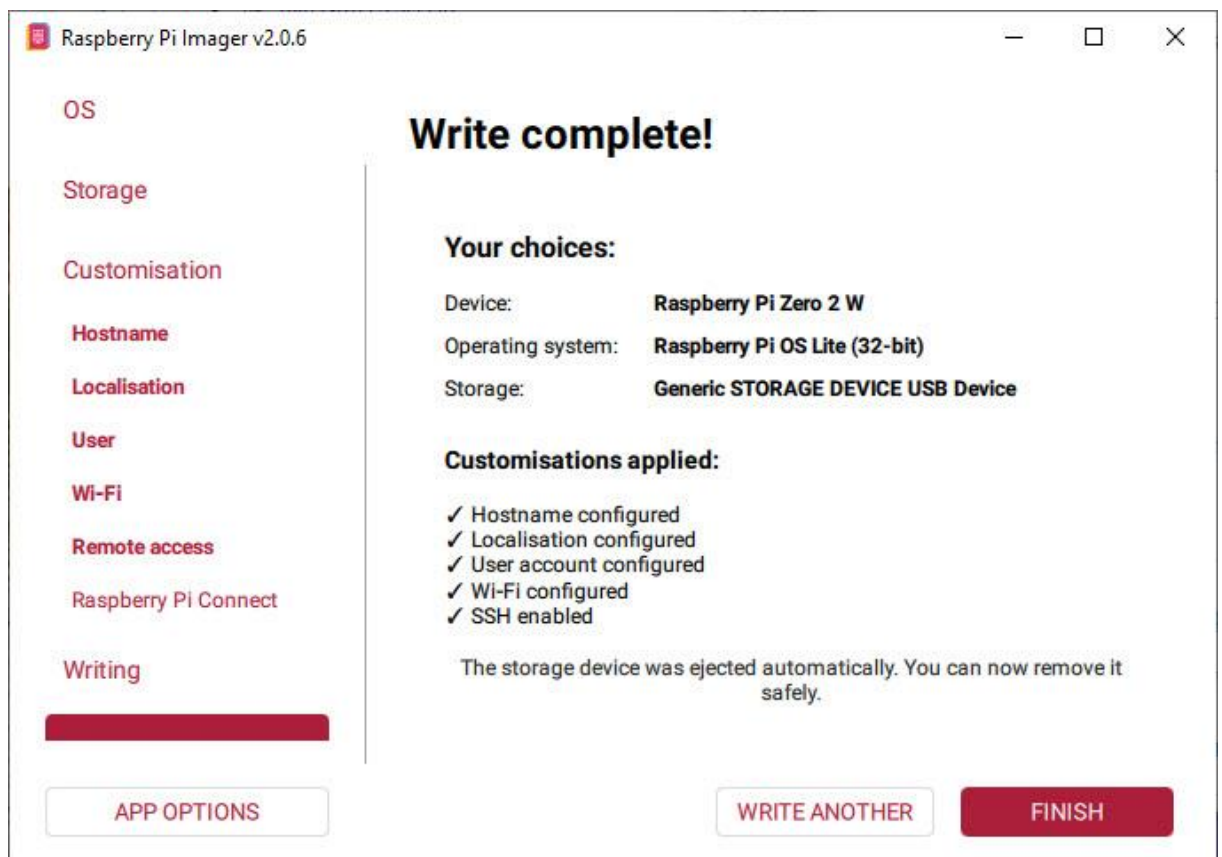
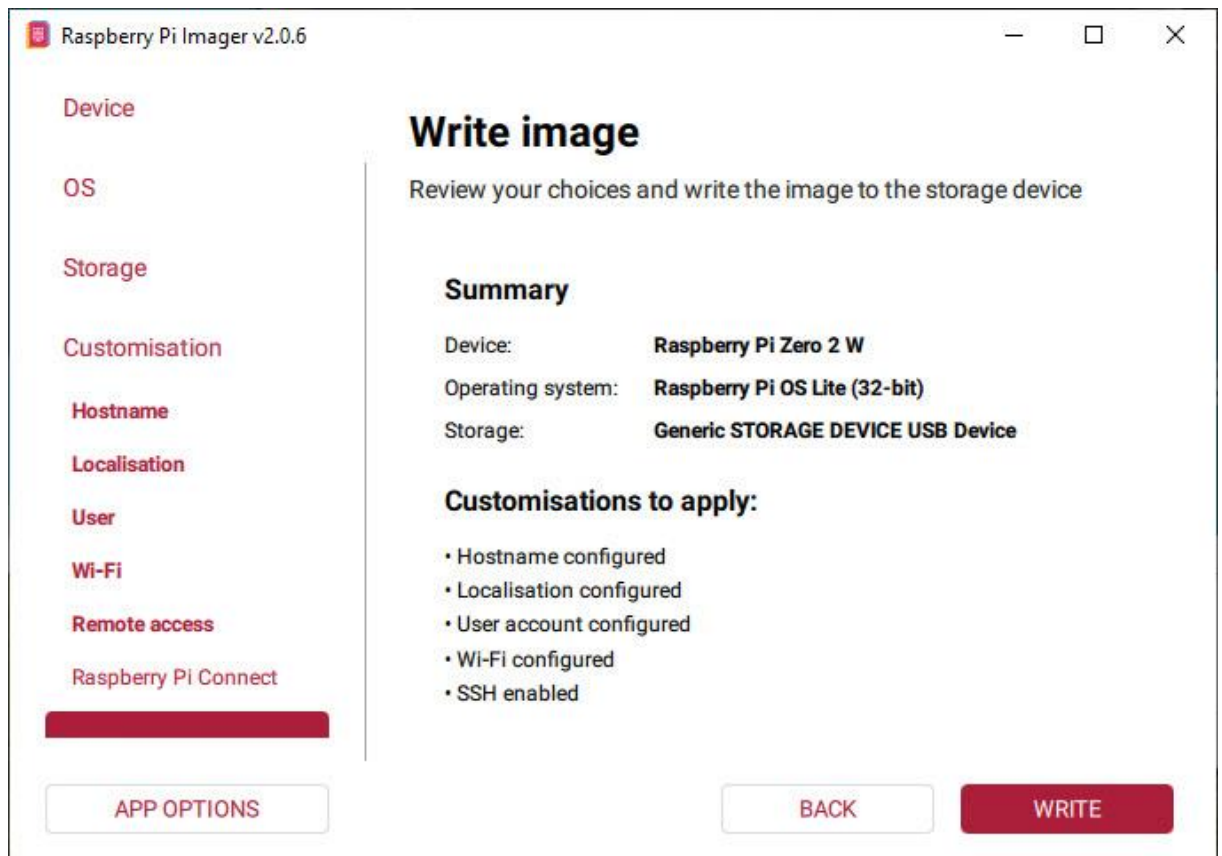
APP OPTIONS

SKIP CUSTOMISATION

BACK

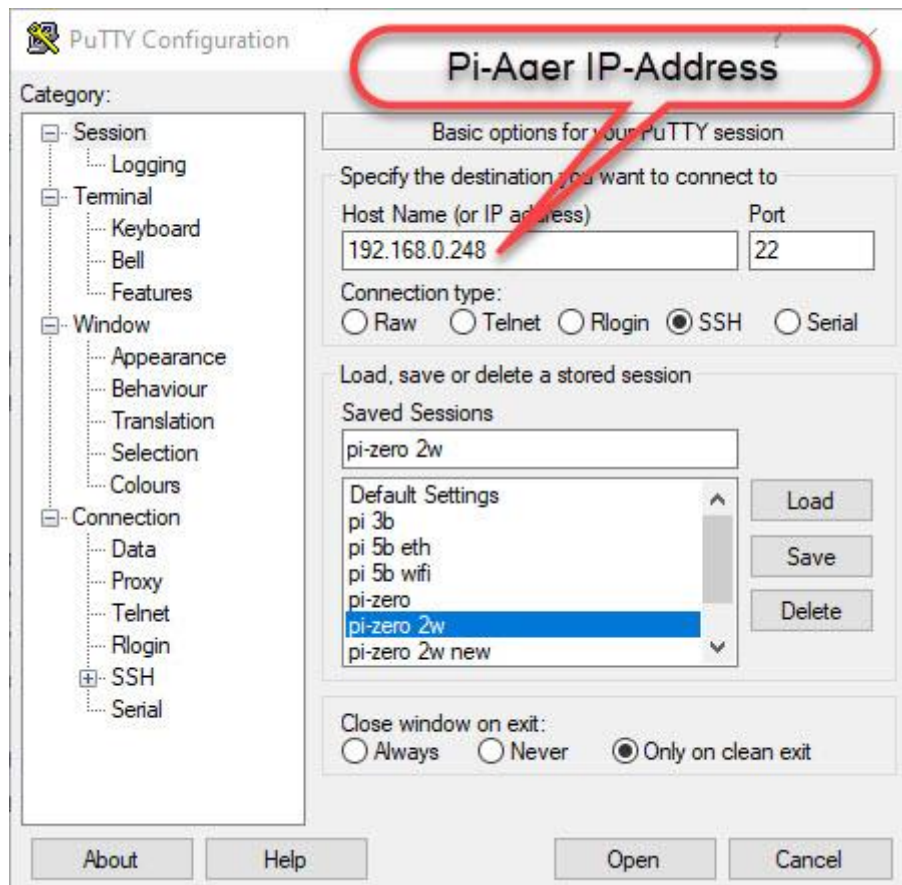
NEXT





Put your SD-Card in your Raspberry Pi and power on your Pi device.
Login via SSH (e.g. use PuTTY) or connect a HDMI monitor and USB keyboard to your Raspberry device and continue with the setup as described below.

- Download and install PuTTY from <https://www.putty.org/>
- Optional download and install WinSCP from <https://winscp.net/eng/index.php>
- Start PuTTY to connect to Pi-Ager:




```
pi@pi-ager: ~  
login as: pi  
pi@192.168.0.206's password:  
Linux pi-ager 6.6.74+rpt-rpi-v7 #1 SMP Raspbian 1:6.6.74-1+rpt1 (2025-01-27) arm  
v7l  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Wed Mar  5 20:00:41 2025 from 192.168.0.201  
  
SSH is enabled and the default password for the 'pi' user has not been changed.  
This is a security risk - please login as the 'pi' user and type 'passwd' to set  
a new password.  
  
pi@pi-ager:~ $
```

- Download system build scripts from github:
`wget -O pi-ager-build-system.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-build-system.sh`
`wget -O pi-ager-finalize-build.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-finalize-build.sh`
`wget -O setup-wifi-ap.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/setup-wifi-ap.sh`

`chmod +x *.sh`

Start now first phase of Pi-Ager System setup:

`sudo ./pi-ager-build-system.sh`

You can now follow the system setup, that is divided in 3 steps with rebooting between the steps. Restart the script after rebooting to continue the system setup :

`sudo ./pi-ager-build-system.sh`

After step 2 you are informed to edit the file `/boot/firmware/setup.txt` that contains information about passwords, WLAN setup parameter, temperature/humidity sensor to use and optional HMI display version connected to your Pi-Ager:

`sudo nano /boot/firmware/setup.txt`

```
pi@pi-ager: ~
GNU nano 7.2 /boot/firmware/setup.txt
# ACHTUNG
# Im Router bzw bei den WLAN Security Einstellungen sollte ein Verschlüsselungsverfahren
# WPA2 (CCMP) oder WPA2+WPA3 ausgewählt werden, da sonst keine WLAN Verbindung zwischen
# Pi-Ager und Router bzw. Accesspoint aufgebaut werden kann.
# WPA wird nicht vom Pi-Ager unterstützt
#-----

# WLAN Netzwerkname:
wlanssid=' '
# WLAN Schlüssel (mindesten 8 Zeichen):
wlankey=' '
# ISO code your country, e.g. DE, GB:
country='DE'

# Temperature/Humidity sensor attached to pi-ager:
# Supported types are : SHT75, SHT85, SHT3x, SHT3x-mod, AHT1x, AHT1x-mod, AHT2x, AHT30, SHT4x-A, S
sensor='SHT75'

# hmi TFT display device Namen:
# gültige device Namen: NX3224K028 or NX3224T028 or NX3224F028
# z.B. : hmidisplay='NX3224K028'
# Wenn hmidisplay definiert ist, wird die Firmware des hmi TFT Displays beim ersten Booten geflash
# Das Flashen des Displays dauert bis zu 5 Minuten, der Fortschritt des Flashens kann nur auf dem
# TFT-Display beobachtet werden.
# Während des Flashens sollte die Stromversorgung des Raspi nicht ausgeschaltet werden.
#
hmidisplay=

# setup.txt nach boot löschen? leer lassen für löschen oder 1 für behalten
keepconf=1

```

After saving setup.txt the last step initializes the Pi-Ager system with data from setup.txt by starting the script pi-ager-finalize-build.sh:

sudo ./pi-ager-finalize-build.sh

The system is then rebooted and ready to start the Pi-Ager. Open your browser window and enter the IP-Address of the Pi-Ager in the address line.

How to generate and deploy Pi-Ager Images

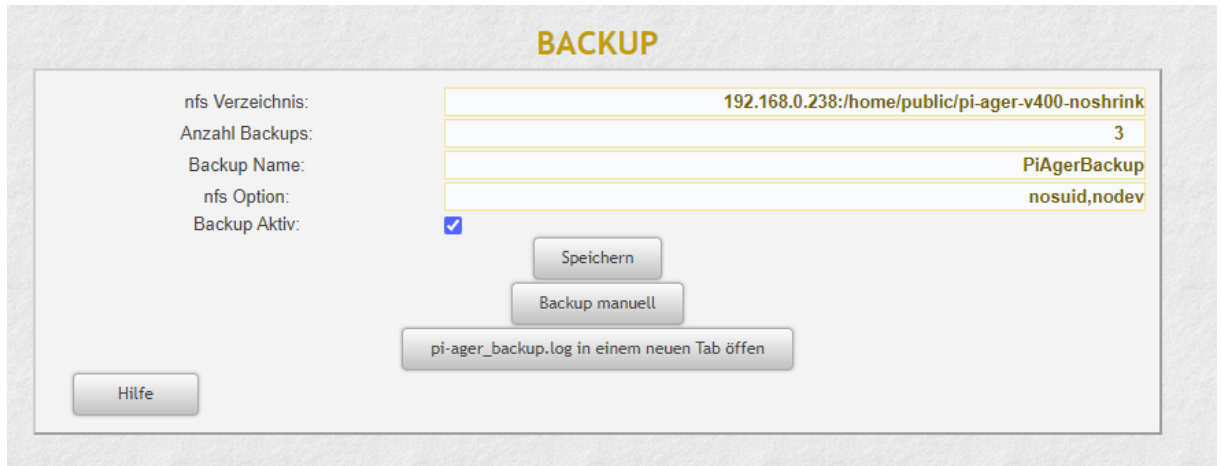
There are some steps to do for generating and deploying a new image from a running Pi-Ager system:

1. Setup a NFS Server to store a new image. The best way is using a Raspi 4B or 5 equipped with an USB Memory stick with 128GB or more. Using RPi 5 it is now possible to add a NVME SSD so that there will be enough space for storing one or more images.

How to setup a NFS Server :

<https://www.elektronik-kompendium.de/sites/raspberry-pi/2007061.htm>

2. On the Pi-Ager setup and start an Image backup on the ADMIN page. For example :



The screenshot shows the 'BACKUP' configuration page in the Pi-Ager ADMIN interface. It features a form with several input fields and buttons. The fields are labeled on the left: 'nfs Verzeichnis:', 'Anzahl Backups:', 'Backup Name:', 'nfs Option:', and 'Backup Aktiv:'. The corresponding values are entered in the fields on the right: '192.168.0.238:/home/public/pi-ager-v400-noshrink', '3', 'PiAgerBackup', and 'nosuid,nodev'. The 'Backup Aktiv:' checkbox is checked. Below the form, there are three buttons: 'Speichern', 'Backup manuell', and 'pi-ager_backup.log in einem neuen Tab öffnen'. A 'Hilfe' button is located at the bottom left of the form area.

nfs Verzeichnis:	192.168.0.238:/home/public/pi-ager-v400-noshrink
Anzahl Backups:	3
Backup Name:	PiAgerBackup
nfs Option:	nosuid,nodev
Backup Aktiv:	☒

Buttons:
Speichern
Backup manuell
pi-ager_backup.log in einem neuen Tab öffnen
Hilfe

3. Next start a terminal on the Pi-Ager and start the pi-ager_image.sh script with option -l: (lowercase 'L')

```
pi@pi-ager: /usr/local/bin

pi@pi-ager:~ $ cd /usr/local/bin/
pi@pi-ager:/usr/local/bin $ sudo pi-ager_image.sh -l
Backup ist inaktiv. Pi-Ager_image wird gestartet!
überprüfe ob der NFS-Server vorhanden ist.
Checking...
192.168.0.238:/home/public/pi-ager-v400-noshrink ist vorhanden
überprüfe ob Backup Pfad vorhanden ist.
Checking ...
/home/nfs/public ist vorhanden
überprüfe ob NFS Mount point definiert ist.
Checking ...
/home/nfs/public ist definiert
überprüfe ob NFS Mount point im file system angelegt ist.
Checking...
/home/nfs/public ist angelegt
NFSVOL=192.168.0.238:/home/public/pi-ager-v400-noshrink
NFSPATH=/home/nfs/public
NFSMOUNT=/home/nfs/public
umount: /home/nfs/public: not mounted.
hänge NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink ein
mount with options: nosuid,nodev
mount NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink successfull. Image script continues
.
Load last backup file.
Backup path is /home/nfs/public
Backup file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Backup path with file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Source File = /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
do_copy = false
my_image = false
Using /home/nfs/public/PiAgerBackup_2024-02-12-103529.img as source and target
#####
parted_output = 2:541065216B:15931539455B:15390474240B:ext4::;
partnum = 2
partstart = 541065216
loopback = /dev/loop0
#####
parted_output_boot = 1:4194304B:541065215B:536870912B:fat32::lba;
partnum_boot = 1
partstart_boot = 4194304
loopback_boot = /dev/loop1
#####
mount directory is /tmp/tmp.SfM3akt9CW
2024-02-12 10:39:10 URL:https://raw.githubusercontent.com/Tronje-the-Falconer/Pi-Ager/entwicklung/boo
t/firmware/setup.txt [3281/3281] -> "setup.txt" [1]
setup.txt copied to /tmp/tmp.SfM3akt9CW/boot/
cmdline.txt modified, added init=/usr/lib/raspberrypi-sys-mods/firstboot
rematch1 /tmp/
rematch2 tmp.SfM3akt9CW
change rootdir is tmp.SfM3akt9CW
/tmp
pi-ager_main.service disabled, will be enabled during setup_pi-ager.sh
Created symlink /etc/systemd/system/multi-user.target.wants/setup_pi-ager.service -> /lib/systemd/system/setup_p
i-ager.service.
setup_pi-ager service enabled
Running in chroot, ignoring command 'is-active'
hostname changed to pi-ager
unmount /tmp/tmp.SfM3akt9CW/boot
unmount /tmp/tmp.SfM3akt9CW
Detaching loop devices from /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
The image /home/nfs/public/PiAger_image_2024-02-12-103940.img was successfully created.
pi@pi-ager:/usr/local/bin $
```

4. Zip and deploy your new image.